

HIV Antibody Detection by ELISA

Laboratory Report — Indirect Enzyme-Linked Immunosorbent Assay

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Technique category: Immunoassay / Serology

1. Objective

To detect the presence of HIV-specific antibodies in serum samples using an indirect Enzyme-Linked Immunosorbent Assay (ELISA), and to interpret plate results against an assay cut-off derived from negative control readings.

2. Introduction

ELISA is a widely used immunoassay technique for detecting and quantifying antigens or antibodies in a biological sample through antibody–antigen binding coupled to an enzyme reporter system. In the indirect format applied here, HIV antigens (recombinant viral proteins such as gp41, gp120, and p24) are immobilised on the surface of a microplate well. If HIV-specific antibodies are present in a patient's serum, they bind to the immobilised antigen. A horseradish peroxidase (HRP)-conjugated anti-human IgG secondary antibody is then added, which binds the captured patient antibody. Addition of a chromogenic substrate (TMB) produces a colour change proportional to the amount of bound antibody, measured spectrophotometrically as optical density (OD) at 450 nm.

3. Materials

- 96-well ELISA microplate pre-coated with HIV antigen
- Patient serum samples, positive control, and negative control sera
- Wash buffer (PBS with 0.05% Tween-20)
- Blocking buffer (PBS with 1–5% BSA)
- HRP-conjugated anti-human IgG secondary antibody
- TMB substrate solution and stop solution (2M H₂SO₄)
- Microplate reader (450 nm filter), micropipettes, 37°C incubator

4. Method

- **Coating:** HIV antigen bound to microplate wells (overnight, 4°C, if not pre-coated).
- **Washing:** Plate washed 3× to remove unbound antigen.
- **Blocking:** Non-specific sites blocked with BSA buffer (1 hr, 37°C).
- **Sample incubation:** Diluted patient serum and controls added to wells (1 hr, 37°C).
- **Secondary antibody:** HRP-conjugated anti-human IgG added (30–60 min, 37°C).
- **Substrate development:** TMB added; colour develops over 10–15 minutes in the dark.
- **Stop & read:** Reaction stopped with acid; absorbance read at 450 nm within 30 minutes.

5. Results

Raw OD450 readings were recorded for two negative controls, two positive controls, and eight patient samples (full dataset in Appendix A). The assay cut-off was calculated as the mean OD450 of the negative controls plus a fixed factor of 0.10, giving a cut-off value of **0.24**. Each patient sample was classified as Non-reactive, Reactive, or Equivocal relative to this cut-off (with a ± 0.02 equivocal margin).

Sample	OD450	Classification
Patient_1	0.112	Non-reactive
Patient_2	0.891	Reactive
Patient_3	0.238	Equivocal
Patient_4	0.098	Non-reactive
Patient_5	1.203	Reactive
Patient_6	0.156	Non-reactive
Patient_7	0.244	Equivocal
Patient_8	0.077	Non-reactive

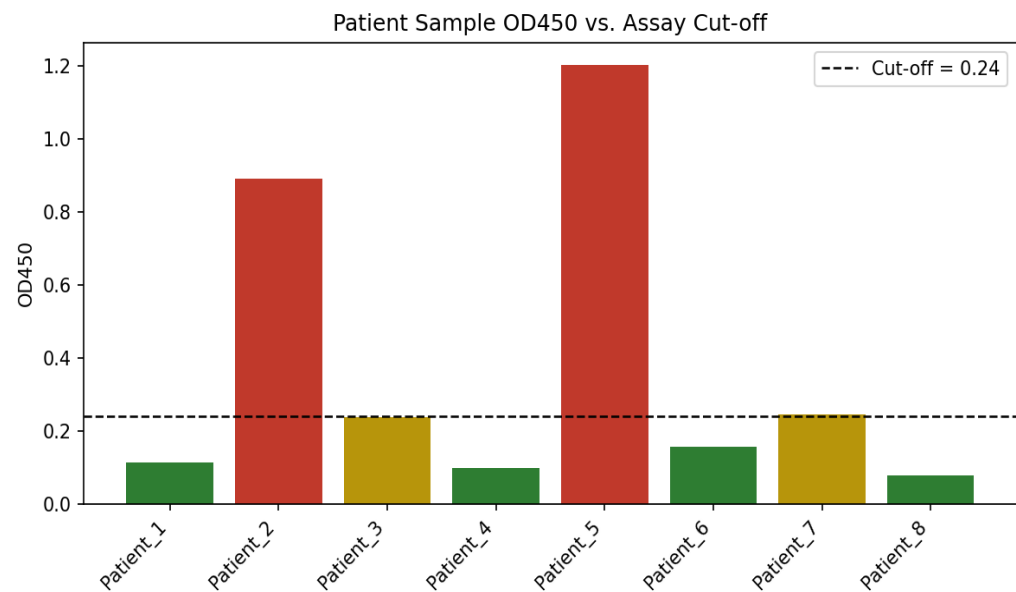


Figure 1. Patient sample OD450 readings relative to the calculated assay cut-off (dashed line).

Of the eight patient samples tested, five were Non-reactive, two were Reactive, and one returned an Equivocal result close to the cut-off threshold.

6. Discussion

The two samples classified as Reactive (Patient_2, Patient_5) showed OD450 values well above the cut-off and close to the positive control range, consistent with the presence of HIV-specific antibodies. The Equivocal samples (Patient_3, Patient_7) fell within the ± 0.02 margin around the cut-off, illustrating why a single ELISA reading near the threshold cannot be treated as conclusive. In practice, both Reactive and Equivocal results would proceed to confirmatory testing (e.g. Western blot or a second ELISA based on a different antigen)

before any diagnostic conclusion is reported, per standard testing algorithms. Control performance (positive control OD >> cut-off; negative controls tightly clustered at low OD) supports the reliability of this plate run.

7. Conclusion

This exercise demonstrated the indirect ELISA workflow for HIV antibody screening end to end — from raw absorbance data to a cut-off-based classification — and reinforced why ELISA is used strictly as a screening tool, with reactive results always requiring confirmatory follow-up.